

$$1) \frac{dy}{dx} = 3x - 1$$

$$dy = (3x - 1)dx$$

$$y = \frac{3x^2}{2} - x + C$$

$$2) ydx - xdy = 0$$

$$ydx = xdy$$

$$\frac{dx}{x} = \frac{dy}{y}$$

$$\ln x = \ln y + \ln k$$

$$\frac{x}{y} = k$$

$$3) \frac{dy}{dx} + y \cos x = 0$$

$$\frac{dy}{dx} = -y \cos x$$

$$\frac{dy}{y} = -\cos x dx$$

$$\ln y = -\sin x + C$$

$$y = e^{-\sin x + C}$$

$$4) (x - 1)dy - ydx = 0$$

$$(x - 1)dy = ydx$$

$$\frac{dy}{y} = \frac{dx}{(x - 1)}$$

$$\ln y = \ln(x - 1) + \ln k$$

$$y = (x - 1)k$$

$$5) (4y + yx^2)dy - (2x + xy^2)dx = 0$$

$$y(4+x^2)dy = x(2+y^2)dx$$

$$\frac{ydy}{2+y^2} = \frac{x dx}{4+x^2}$$

$$\frac{1}{2} \ln|2+y^2| = \frac{1}{2} \ln|4+x^2| + \ln|k|$$

$$\ln|2+y^2| = \ln|4+x^2| + \ln|k|$$

$$2+y^2 = (4+x^2)C$$

$$6) \sin x(e^{-y} + 1)dx = (1 + \cos x)dy; x = 0, y = 0$$

$$\frac{\sin x dx}{(1 + \cos x)} = \frac{dy}{e^y + 1}$$

$$\frac{\sin x dx}{(1 + \cos x)} = \frac{dy}{\frac{1}{e^y} + 1}$$

$$\frac{\sin x dx}{1 + \cos x} = \frac{e^y dy}{1 + e^y}$$

$$\frac{dy}{\frac{1}{1+e^y}}$$

$$\ln|k| - \ln|1 + \cos x| = \ln|1 + e^y|$$

$$\ln|k| = \ln|1 + e^y| + \ln|1 + \cos x|$$

$$k = (1 + e^y)(1 + \cos x)$$

$$k = (1 + 1)(1 + 1)$$

$$k = 4$$

$$(1 + e^y)(1 + \cos x) = 4$$

$$7) ydy = 4x\sqrt{y^2 + 1} dx; x = 0, y = 1$$

$$\frac{ydy}{\sqrt{y^2 + 1}} = 4x dx$$

$$t = y^2 + 1$$

$$dt = 2y dy$$

$$\sqrt{y^2 + 1} = 2x^2 + C$$

$$\sqrt{2} = 0 + C$$

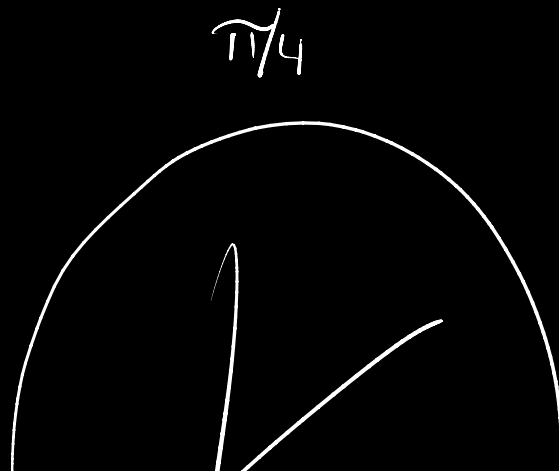
$$C = \sqrt{2}$$

$$\sqrt{y^2 + 1} = 2x^2 + \sqrt{2}$$

$$8) \frac{dx}{dy} = 4(x^2 + 1); x = 1, y = \frac{\pi}{4}$$

$$dx = 4du$$

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$$\frac{1}{x^2+1} = \frac{1}{2} \cdot \frac{2}{x^2+1}$$

$$\arctan(x) = \frac{1}{2} \pi + C$$

$$\frac{\pi}{4} = \pi + C$$

$$-\frac{3}{4}\pi = C$$

$$\arctan(x) = \frac{1}{2} \pi - \frac{3}{4}\pi$$

$$x = \tan\left(\frac{1}{2}\pi - \frac{3}{4}\pi\right)$$

$$9) e^x y \frac{dy}{dx} = e^{-y} + e^{-2x-y}$$

$$e^x y \frac{dy}{dx} = \frac{1}{e^y} + \frac{1}{e^{2x} \cdot e^y}$$

$$e^x y \frac{dy}{dx} = \frac{e^{2x} + 1}{e^y e^{2x}}$$

$$e^y y dy = \frac{(e^{2x} + 1)}{e^{2x}} dx$$

$$e^y y dy = \left(e^x + \frac{1}{e^x}\right) dx$$

$$e^y y - y = e^x - \frac{1}{e^x} + C$$

$$\int u dv = u \cdot v - \int v du$$

$$u = y = e^y y - \int e^y dy$$

$$dv = e^y = e^y y - e^y$$

$$v = e^y$$